Auteur: Karanjot Singh
Studierichting: Informatica
Oefeningenreeks 6A nr. 6.15

$$\int_0^1 \frac{dx}{x^2+4}$$

We gebruiken de substitutie $x = 2\tan\theta$, dus $dx = 2\sec^2\theta\, d\theta$

Pas de grenzen aan:

Als $x = 0$, dan $\theta = 0$

Als $x = 1$, dan $1 = 2\tan\theta \Rightarrow \theta = \text{Bgtan}\left(\frac{1}{2}\right)$

Vervang in de integrand:

$$\int_0^1 \frac{dx}{x^2+4} = \int_0^{\text{Bgtan}\left(\frac{1}{2}\right)} \frac{2\sec^2\theta\, d\theta}{4\tan^2\theta+4}$$
$$= \int_0^{\text{Bgtan}\left(\frac{1}{2}\right)} \frac{2\sec^2\theta\, d\theta}{4(\tan^2\theta+1)}$$
$$= \int_0^{\text{Bgtan}\left(\frac{1}{2}\right)} \frac{2\sec^2\theta\, d\theta}{4\sec^2\theta}$$
$$= \int_0^{\text{Bgtan}\left(\frac{1}{2}\right)} \frac{2}{4}\, d\theta = \frac{1}{2}\int_0^{\text{Bgtan}\left(\frac{1}{2}\right)} d\theta$$
$$= \frac{1}{2}\left(\text{Bgtan}\left(\frac{1}{2}\right) - 0\right)$$
$$= \frac{1}{2}\,\text{Bgtan}\left(\frac{1}{2}\right)$$

Dus, $\displaystyle\int_0^1 \frac{dx}{x^2+4} = \frac{1}{2}\,\text{Bgtan}\left(\frac{1}{2}\right)$

References